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Pushbutton switch, pushbutton switch unit, operating unit, and amusement machine

Abstract

A pushbutton switch according to one or more embodiments includes an attachment fixed to one surface of a support, an operable portion removably engaged with the attachment, and a detector attached to another surface of the support. The attachment has a loose-fitting hole to loosely receive the movable magnet with the operable portion being engaged with the attachment.

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Background/Summary

FIELD

(1) The present disclosure relates to a pushbutton switch, a pushbutton switch unit, an operating unit, and an amusement machine.

BACKGROUND

(2) Patent Literature 1 describes a known pushbutton assembly. The pushbutton assembly described in Patent Literature 1 includes a display and at least one pushbutton switch. The display includes a transparent material on its display surface to support the pushbutton switch. The transparent material is a support for the pushbutton switch.

CITATION LIST

Patent Literature

(3) Patent Literature 1: U.S. Pat. No. 10,431,037

SUMMARY

Technical Problem

(4) In the pushbutton assembly described in Patent Literature 1, the pushbutton switch is received in an attachment opening in the transparent material. However, the structure with an opening may lower the strength of the transparent material. The pushbutton switch may be included in a machine such as an amusement machine and operated by a user with strong force. In this case, in particular, the transparent material may fracture with cracks extending from the opening.

(5) A pushbutton switch according to one or more embodiments includes an operable portion attached to one surface of a transparent material and a detector attached to the other surface of the transparent material. The operable portion includes a press portion to be operated by a user. The detector detects the pressed state of the press portion. The pushbutton switch may be installable on the transparent material without an attachment opening. The pushbutton switch includes consumable components, specifically the press portion or other moving components, or

components that support these components. Such components are to be replaceable.

(6) The pushbutton switch thus may allow easy attachment and removal of the operable portion. The operable portion may be attachable and removable at the front surface of the transparent material to facilitate replacement. In this case, however, the operable portion is to be prevented from being removed by the user with, for example, ill intentions.

(7) One or more embodiments is directed to a technique for preventing an operable portion from being removed from a pushbutton switch by a person other than the manager.

(8) A switch, a unit, and a machine according to one or more embodiments have the structures described below. A pushbutton switch according to one or more embodiments includes an attachment fixed to one surface of a support being a plate, an operable portion removably engaged with the attachment, and a detector attached to another surface of the support and facing the operable portion. The operable portion includes a press portion and a movable magnet movable in a pressing direction from the operable portion toward the detector and in a direction opposite to the pressing direction. The operable portion is rotatable about an axis aligned with the pressing direction to be engaged with the attachment. The attachment has a loose-fitting hole to loosely receive the movable magnet with the operable portion being engaged with the attachment.

(9) An operating unit according to one or more embodiments includes a transparent support and a display together serving as a touchscreen, and a pushbutton switch. The pushbutton switch includes an attachment fixed to a surface of the transparent support opposite to the display, an operable portion removably engaged with the attachment, and a detector attached to a surface of the transparent support adjacent to the display. The detector faces the operable portion. The operable portion includes a press portion and a movable magnet movable in a pressing direction from the operable portion toward the detector and in a direction opposite to the pressing direction. The operable portion is rotatable about an axis aligned with the pressing direction to be engaged with the attachment. The attachment has a loose-fitting hole to loosely receive the movable magnet with the operable portion being engaged with the attachment.

(10) An amusement machine according to one or more embodiments includes a first display that displays an image for amusement, a transparent support and a second display together serving as a touchscreen, and a pushbutton switch. The pushbutton switch includes an attachment fixed to a surface of the transparent support opposite to the second display, an operable portion removably engaged with the attachment, and a detector attached to a surface of the transparent support adjacent to the second display. The detector faces the operable portion. The operable portion includes a press portion and a movable magnet movable in a pressing direction from the operable portion toward the detector and in a direction opposite to the pressing direction. The operable portion is rotatable about an axis aligned with the pressing direction to be engaged with the attachment. The attachment has a loose-fitting hole to loosely receive the movable magnet with the operable portion being engaged with the attachment.

(11) The technique according to one or more embodiments may prevent the operable portion from being removed from the pushbutton switch by a person other than the manager.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram illustrating an external perspective view of an amusement machine according to a first embodiment or embodiments.

(2) FIG. 2 is a diagram illustrating a plan view of a button deck that is an operating unit of an amusement machine, such as is shown in FIG. 1.

(3) FIG. 3 is a diagram illustrating a cross-sectional view taken along line A-A as viewed in a direction indicated by arrows in FIG. 2.

- (4) FIG. 4 is a diagram illustrating an exploded perspective view of a button deck, such as is shown in FIG. 2.
- (5) FIG. 5 is a diagram illustrating a bottom view of a button deck, such as is shown in FIG. 2.
- (6) FIG. 6 is a diagram illustrating a view of an example display on a button deck, such as is shown in FIG. 2.
- (7) FIG. 7 is a diagram illustrating an exploded perspective view of an operable portion of a pushbutton switch in a button deck, such as is shown in FIG. 2.
- (8) FIG. 8 is a diagram illustrating a perspective view of a base supporting multiple components in an operable portion, such as is shown in FIG. 7.
- (9) FIG. 9 is a diagram illustrating an exploded perspective view of a detector in a pushbutton switch in a button deck, such as is shown in FIG. 2.
- (10) FIG. 10 is a block diagram illustrating an amusement machine, such as is shown in FIG. 1, that includes a control system.
- (11) FIG. 11 is a diagram illustrating views of a structure for detecting a pressed state of a button in a pushbutton switch in a button deck, such as is shown in FIG. 2.
- (12) FIG. 12 is a diagram illustrating a plan view of a first attachment in a button deck, such as is shown in FIG. 4.
- (13) FIG. 13 is a diagram illustrating a cross-sectional view of tabs on a first attachment in a button deck, such as is shown in FIG. 4.
- (14) FIG. 14 is a diagram illustrating cross-sectional views of an engagement portion of a base for engagement with a corresponding tab, such as is shown in FIG. 13.
- (15) FIG. 15 is a diagram illustrating views of a lock assembly for an operable portion, such as is shown in FIG. 7.
- (16) FIG. 16 is a diagram illustrating a perspective view of a button deck, such as is shown in FIG. 2 and a fixture.
- (17) FIG. 17 is a diagram illustrating a perspective view of a button deck, such as is shown in FIG. 2 and a fixture attached to a pushbutton switch.
- (18) FIG. 18 is a diagram illustrating a view showing a lock assembly for an operable portion, such as is shown in FIG. 7 and a fixture unlocking an operable portion.
- (19) FIG. 19 is a diagram illustrating a perspective view of a button deck in a second embodiment or embodiments and a fixture attached to a pushbutton switch.
- (20) FIG. 20 is a diagram illustrating magnets in an operable portion of a pushbutton switch in a button deck, such as is shown in FIG. 19 and magnets in a fixture, showing arrangements.

DETAILED DESCRIPTION

First Embodiment

(21) One or more embodiments of the present invention (hereafter also referred to as the present embodiment) will now be described with reference to the drawings. In the present embodiment, a pushbutton switch and an operating unit according to an aspect of the present invention are included in an amusement machine.

(22) The present invention is not limited to the embodiments described below, and may be variously designed without departing from the spirit and scope of the invention. The pushbutton switch and the operating unit according to an aspect of the present invention can also be used for industrial equipment and consumer equipment, as well as for various amusement machines. 1.

Example Use

(23) As shown in FIG. 1, a pushbutton switch 15 and a button deck 10, which is an operating unit including the pushbutton switch 15, may be included in an amusement machine 1. As shown in FIGS. 3 and 4, the button deck 10 includes the pushbutton switch 15 and a display input unit 11. The display input unit 11 includes a transparent support plate 13 and a display 12 and serves as a touchscreen.

(24) The pushbutton switch 15 includes, as separate portions, an operable portion 20 and a detector

30. The operable portion 20 includes a button (a press portion) 21. The detector 30 detects the pressed state of the button 21. The operable portion 20 is attached to one surface of the transparent support plate 13. The detector 30 is attached to the other surface of the transparent support plate 13. The operable portion 20 and the detector 30 face each other with the transparent support plate 13 in between. The detector 30 includes sensors for detecting the pressed state of the button 21 through the transparent support plate 13 (with the transparent support plate 13 in between).

(25) As shown in FIG. 4, the operable portion 20 is attached to a first attachment 41 fixed to a first surface 13a of the transparent support plate 13. The operable portion 20 is removably engaged with the first attachment 41. As shown in FIG. 7, the operable portion 20 includes a locking magnet 27. The locking magnet 27 is a movable magnet movable in the pressing direction of the button 21 and in the direction opposite to the pressing direction.

(26) As shown in FIGS. 12 and 15, the first attachment 41 has a loose-fitting hole 41h for loosely receiving the locking magnet 27 with the operable portion 20 being engaged with the first attachment 41. When the operable portion 20 is engaged with the first attachment 41, the locking magnet 27 is loosely fitted in the loose-fitting hole 41h to lock the engaged operable portion 20. This prevents the operable portion from being removed by a person other than the manager of the amusement machine 1, such as a player.

(27) 2. Example Structure

(28) 1. Amusement Machine

(29) The schematic structure of the amusement machine according to the present embodiment will now be described with reference to FIG. 1. FIG. 1 is an external perspective view of the amusement machine 1 according to the present embodiment.

(30) As shown in FIG. 1, the amusement machine 1 includes a main display unit (a first display) 2, a button deck (an operating unit) 10, and a housing 3 supporting these components.

(31) The main display unit 2 displays images for a game played on the amusement machine 1 and includes, for example, a liquid crystal display (LCD). The main display unit 2 has its display surface facing the front of the amusement machine 1.

(32) The button deck 10 receives the player's operations in a game on the amusement machine 1. In the present example, the button deck 10 is located below the main display unit 2 at the front of the amusement machine 1. The button deck 10 is rectangular and elongated laterally. The button deck 10 includes the display input unit 11 as a touchscreen, and the pushbutton switch 15.

(33) For the amusement machine 1 providing slot machine gaming involving spinning reels, for example, the main display unit 2 displays multiple reels with multiple types of symbols. The button deck 10 receives the player's instruction for spinning the multiple reels, and receives the number of paylines and the number of bets selected by the player.

(34) The multiple reels start spinning in response to the player's instruction for spinning received by the button deck 10. The reels automatically stop spinning under the control of a main controller 60 (refer to FIG. 10, described later). The pattern is determined by the combination of symbols on paylines and by the symbols appearing when the multiple reels are stopped. The player is rewarded based on the determined pattern and the number of bets.

(35) 2. Button Deck

(36) FIG. 2 is a plan view of the button deck 10. FIG. 3 is a cross-sectional view taken along line A-A as viewed in the direction indicated by arrows in FIG. 2. FIG. 4 is an exploded perspective view of the button deck 10. FIG. 5 is a bottom view of the button deck 10.

(37) As shown in FIG. 2, the display input unit 11 is rectangular and elongated laterally. The display input unit 11 includes the pushbutton switch 15 on its display surface. In the present example, the pushbutton switch 15 is circular and at the right end as viewed in plan (refer to FIG. 1).

(38) As shown in FIG. 3, the display input unit 11 includes a display (a second display) 12 and a transparent support plate (a support) 13 located on the display surface of the display 12. In the

present example, the transparent support plate **13** is a position input device. The transparent support plate **13** and the display **12** together serve as a touchscreen. The display **12** displays images of keys for receiving inputs to the amusement machine **1**, images for a game, or other images.

(39) The pushbutton switch **15** includes the operable portion **20** and the detector **30**. The operable portion **20** is attached to one surface, specifically a first surface **13a**, of the transparent support plate **13**. The first surface **13a** is opposite to the surface facing the display **12**. The operable portion **20** includes the button (a press portion) **21** to receive a pressing operation. The button **21** is transparent.

(40) The detector **30** is attached to the other surface, specifically a second surface **13b**, of the transparent support plate **13** and faces the operable portion **20**. The second surface **13b** is adjacent to the display **12**. The detector **30** includes sensors for detecting the pressed state of the button **21** through the transparent support plate **13**.

(41) In the present example, the sensors are reflective sensors **34** (refer to FIG. 9). However, the sensors may be any sensors other than the reflective sensors **34**, such as range sensors, that can detect the pressed state of the button **21** through the transparent support plate **13**. Electronic components for detecting a pressing operation may be located in the operable portion **20**. Such a structure allows a wider choice of sensors.

(42) In the present example, the detector **30** is formed from a transparent material in its area facing the button **21** to allow the display surface of the display **12** to be viewed through the transparent button **21**. The detector **30** may have an opening in its area facing the button **21**.

(43) In the present example, the pushbutton switch **15** is located above the display **12**. The support is the transparent support plate **13** that is entirely transparent except for its outer edge portion. However, the pushbutton switch **15** may not be located above the display **12**. In this case, the support may be transparent simply in its area receiving the operable portion **20** and the detector **30**, or more specifically in its area through which the sensors in the detector **30** detect the pressed state of the button **21**.

(44) In the present example, as shown in FIG. 4, the pushbutton switch **15** includes the first attachment **41** (an attachment), a second attachment **43**, and a locking member **45**. The operable portion **20** is attached to the first surface **13a** of the transparent support plate **13** with the first attachment **41** in between. The operable portion **20** is removably engaged with the first attachment **41**.

(45) The first attachment **41** has an annular shape corresponding to the shape of a bezel **22** (refer to FIG. 7, described later) in the operable portion **20**. The first attachment **41** is fixed to the first surface **13a** using a double-sided tape piece **42** in the present example. Attaching the operable portion **20** to the first attachment **41** is described later.

(46) The detector **30** is attached to the second surface **13b** of the transparent support plate **13** with the second attachment **43** in between. The detector **30** is removably engaged with the second attachment **43**. In the present example, the second attachment **43** together with the locking member **45** allows attachment of the detector **30** to the transparent support plate **13**. Each of the second attachment **43** and the locking member **45** has an annular shape and is fitted on the outer periphery of the detector **30**. The second attachment **43** receives the detector **30** on its inner periphery, and the locking member **45** is fitted on the outer periphery of the detector **30**.

(47) The locking member **45** is rotated relative to the second attachment **43** and engaged with the second attachment **43**. This attaches the detector **30** to the second surface **13b**. The second attachment **43** is fixed to the second surface **13b** using a double-sided tape piece **44** in the present example. Further, as shown in FIGS. 3 and 5, the display **12** in the display input unit **11** has its back surface connected to a relay board (a controller) **47** (described later). The relay board **47** is connected to the detector **30** through a flexible printed circuit (FPC) **37**.

(48) FIG. 6 is a view showing example display on the button deck **10**. The example display in FIG. 6 shows an operation screen for a slot machine game played on the amusement machine **1**. The

pushbutton switch **15** serves as a SPIN button that receives the instruction for spinning the multiple reels. The display **12** displays the letters SPIN indicating the SPIN button at the position corresponding to the button **21**. The letters SPIN are viewable through the transparent button **21**. The display input unit **11** displays LINE buttons **17** for selecting the number of paylines and BET buttons **18** for selecting the number of bets.

(49) 3. Pushbutton Switch Structure

(50) 3-1. Operable Portion

(51) The operable portion **20** will now be described with reference to FIGS. **3**, **7**, and **8**. FIG. **7** is an exploded perspective view of the operable portion **20** of the pushbutton switch **15** in the button deck **10**. FIG. **8** is a perspective view of a base **23** supporting multiple components in the operable portion **20**.

(52) As shown in FIG. **7**, the operable portion **20** includes the button **21**, the bezel **22**, the base **23**, multiple springs **24**, multiple tactile-sensation rubber pieces **25**, the locking magnet **27**, and multiple reflective flappers **26**. The button **21** is supported between the bezel **22** and the base **23** and can receive a pressing operation. In the pushbutton switch **15**, the operable portion **20** is located substantially horizontally, and the button **21** has the pressing direction being downward and the returning direction being upward.

(53) The button **21** is formed from a transparent material and is circular in the present example (refer to FIG. **2**). The button **21** includes a flange **21a** on its outer periphery. The button **21** can be pressed with the flange **21a** being held between the base **23** and the bezel **22** attached to the base **23**. The bezel **22** has an opening **22a** through which an upper surface **21b** of the button **21** is exposed.

(54) The base **23** supports, for example, the button **21**, the bezel **22**, the multiple springs **24**, the multiple tactile-sensation rubber pieces **25**, the locking magnet **27**, and the multiple reflective flappers **26**. The base **23** is fastened to the transparent support plate **13** with the first attachment **41** in between. The base **23** is rotatable about an axis aligned with the pressing direction from the operable portion **20** toward the detector **30** for engagement with the first attachment **41**. The base **23** being engaged with the first attachment **41** can tightly fasten the operable portion **20** to the first attachment **41**.

(55) As shown in FIG. **8**, the base **23** includes an annular component compartment **23a** in an outer peripheral portion. The component compartment **23a** accommodates the multiple springs **24**, the multiple tactile-sensation rubber pieces **25**, the locking magnet **27**, and the multiple reflective flappers **26**. The multiple springs **24** return the pressed button **21** to the position before pressing. The multiple tactile-sensation rubber pieces **25** provide a tactile sensation to the player pressing the button **21**. In the present example, three springs **24** and three tactile-sensation rubber pieces **25** are included.

(56) The multiple springs **24** and the multiple tactile-sensation rubber pieces **25** are in contact with the flange **21a** on the outer periphery of the button **21** (refer to FIG. **3**). The multiple springs **24** in contact with the flange **21a** can return the pressed button **21** to its original position. The multiple tactile-sensation rubber pieces **25** in contact with the flange **21a** can provide a tactile sensation to the player pressing the button **21**.

(57) The reflective flappers **26** are used when the detector **30** detects the pressed state of the button **21**. In the present example, three (multiple) reflective flappers **26** are included. The reflective flappers **26** are supported pivotally on support portions **23b** of the component compartment **23a**. Each reflective flapper **26** includes a shaft **26a** extending in the circumferential direction of the component compartment **23a**.

(58) Each reflective flapper **26** includes a press piece **26b** located inward from the shaft **26a** and a reflective portion **26c** located outward from the shaft **26a**. The press piece **26b** comes in contact with and is pressed by a lower end **21d** (refer to views **1000** and **1001** in FIG. **11**) of an outer peripheral portion of the button **21**. The reflective portion **26c** has its lower surface being a

reflective surface. The reflective flapper **26** changes its orientation in accordance with the pressed or unpressed state of the button **21** and thus changes the orientation of the reflective surface (refer to the views **1000** and **1001** in FIG. **11**).

(59) The component compartment **23a** has openings at the bottom to allow passage of light from the reflective sensors **34** in the detector **30** and allow passage of light reflected by the reflective flappers **26** toward the reflective sensors **34**. The first attachment **41** also has such openings, specifically openings **41t** (refer to FIG. **12**).

(60) The locking magnet **27** is rod-like and has the direction of magnetization parallel to the pressing direction of the button **21**. The locking magnet **27** is a movable magnet movable in the pressing direction of the button **21** and in the direction opposite to the pressing direction. The locking magnet **27** is loosely fitted in the loose-fitting hole **41h** (refer to FIG. **15**) in the first attachment **41** to lock the operable portion **20** and prevent removal from the first attachment **41**.

(61) The locking magnet **27** is fitted loosely in a hole **23c** in the bottom of the component compartment **23a**. The locking magnet **27** includes a head with a larger diameter than the other portion. The locking magnet **27** is prevented from falling out of the hole **23c** with its head stuck in the hole **23c**.

(62) The component compartment **23a** includes an inner peripheral wall **23d** defining the component compartment **23a**. The wall **23d** guides the button **21** to move vertically when the button **21** is pressed or returns. As shown in FIG. **3**, the button **21** has a groove **21c** on its lower surface to receive the upper end of the wall **23d**.

(63) To prepare the operable portion **20**, the button **21** and the bezel **22** are fitted from above in this order onto the component compartment **23a** accommodating the above various components. The base **23** and the bezel **22** are then fastened together with multiple screws **28**.

(64) 3-2. Detector

(65) The detector **30** will now be described with reference to FIGS. **3** and **9**. FIG. **9** is an exploded perspective view of the detector **30** in the pushbutton switch **15** in the button deck **10**. As shown in FIG. **9**, the detector **30** includes a case **31**, a mounting board **32**, a lower cover **33**, the multiple reflective sensors **34**, and the FPC **37**.

(66) The multiple reflective sensors **34** are mounted on the mounting board **32**. In the present example, three reflective sensors **34** are included to correspond to the three reflective flappers **26** in the operable portion **20**. Each reflective sensor **34** faces the corresponding reflective flapper **26** with the operable portion **20** and the detector **30** being attached to the transparent support plate **13**.

(67) The reflective sensor **34** is an optical sensor including a light emitter **34a** and a light receiver **34b**. The light emitter **34a** emits light toward the reflective flapper **26** through the transparent support plate **13**. The reflective flapper **26** reflects the light that is then received by the light receiver **34b**. The light receiver **34b** receives light with intensity changeable in accordance with the orientation of the reflective flapper **26**. The change in light intensity thus allows detection of the pressed state of the button **21**.

(68) As shown in FIG. **9**, the mounting board **32** incorporates multiple light-emitting diodes (LEDs) (light emitters or electronic components) **38** as well as the multiple reflective sensors **34**. The detector **30** can thus emit light when the multiple LEDs **38** is on. The FPC **37** has one end connected to the mounting board **32** and the other end connected to the relay board **47** (refer to FIG. **3**) attached to the back surface of the display **12**. The FPC **37** electrically connects the mounting board **32** and the relay board **47**.

(69) The case **31** supports, for example, the mounting board **32** and the multiple reflective sensors **34**. The lower cover **33** is fitted to the case **31** and covers the bottom of the detector **30**. To prepare the detector **30**, the lower cover **33** is fitted onto the bottom of the case **31** accommodating, for example, the mounting board **32** and the multiple reflective sensors **34**. The case **31** and the lower cover **33** are then fastened together with multiple screws **39**.

(70) 4. Control System in Amusement Machine

(71) FIG. 10 is a block diagram of the amusement machine 1, showing a control system. As shown in FIG. 10, the amusement machine 1 includes the main controller 60 and the relay board 47 as a controller. The main controller 60 is connected to the main display unit 2 and causes the main display unit 2 to display images for a game to perform the game.

(72) The main controller 60 is also connected to the relay board 47 and receives, using the relay board 47, an instruction input on the button deck 10. The main controller 60 also controls the button deck 10 using the relay board 47. The relay board 47 controls the operation of the button deck 10, and is connected to the detector 30 through the FPC 37 and also connected to the display input unit 11 as a touch screen.

(73) As described above, the button deck 10 in the present example includes the pushbutton switch 15 including, as separate portions, the operable portion 20 and the detector 30. The electronic components are located in the detector 30 alone attached to the second surface 13b of the transparent support plate 13 adjacent to the inside of the machine. The relay board 47 is thus unconnected to the operable portion 20.

(74) 5. Detecting Pressing Operation with Pushbutton Switch

(75) FIG. 11 includes views each showing a structure for detecting the pressed state of the button 21 in the pushbutton switch 15 in the button deck 10. FIG. 11 includes a cross-sectional view 1000 of the main part of the pushbutton switch 15 with the button 21 being unpressed, and includes a cross-sectional view 1001 of the main part of the pushbutton switch 15 with the button 21 being pressed. FIG. 11 also includes a cross-sectional view 1002 of the main part of the pushbutton switch 15 being unpressed. The view 1002 is taken along a line different from the line for the view 1001.

(76) As shown in the views 1000 and 1002 in FIG. 11, when the button 21 is unpressed, each reflective flapper 26 in the operable portion 20 is in a first orientation to reflect light from the corresponding reflective sensor 34 in the detector 30 in the direction back toward the reflective sensor 34. In the first orientation, the reflective surface reflects light from the light emitter 34a in the direction for entry into the light receiver 34b.

(77) The light receiver 34b receives, through the transparent support plate 13, light L emitted from the light emitter 34a and reflected by the reflective flapper 26 in the first orientation. The light receiver 34b outputs an electrical signal corresponding to the intensity of the received reflected light, and allows detection of the button 21 being unpressed based on the electrical signal.

(78) As shown in the view 1001 in FIG. 11, when the button 21 is pressed, each reflective flapper 26 in the operable portion 20 is in a second orientation to reflect light from the corresponding reflective sensor 34 in the detector 30 in a direction different from the direction toward the reflective sensor 34. In the second orientation, the reflective surface reflects light from the light emitter 34a in a direction different from the direction for entry into the light receiver 34b.

(79) The light receiver 34b does not receive light L emitted from the light emitter 34a and reflected by the reflective flapper 26 in the second orientation, and thus receives light with reduced intensity. The light receiver 34b outputs a lower electrical signal in accordance with the reduced intensity of received light. This change in the electrical signal allows detection of the button 21 being pressed.

(80) 6. Tabs 41a on First Attachment 41

(81) FIG. 12 is a plan view of the first attachment in the button deck shown in FIG. 4. FIG. 13 is a cross-sectional view showing tabs 41a on the first attachment 41 in the button deck 10 shown in FIG. 4. FIG. 14 includes cross-sectional views each showing an engagement portion 20a of the base 23 for engagement with the corresponding tab 41a shown in FIG. 13.

(82) As shown in FIGS. 12 and 13, the first attachment 41 includes the multiple tabs 41a protruding upward. Each tab 41a is an L-shaped hook. As shown in FIG. 14, the base 23 includes the engagement portion 20a engageable with the tab 41a. The engagement portion 20a is circumferentially adjacent to an arc-shaped hole in the bottom surface of the base 23 extending along the outer edge of the base 23.

(83) The engagement portion **20a** of the base **23** in the operable portion **20** is engaged with the tab **41a** on the first attachment **41** to engage the operable portion **20** with the first attachment **41**. The base **23** in the operable portion **20** includes multiple engagement portions **20a** corresponding to the multiple tabs **41a**.

(84) In the state shown in a view **1101** in FIG. **14**, the tab **41a** is spaced from the engagement portion **20a**. When the base **23** is rotated to engage with the first attachment **41**, the engagement portion **20a** engages with the tab **41a** as shown in a view **1102** in FIG. **14**. The engagement portion **20a** being engaged with the tab **41a** can securely engage the operable portion **20** with the first attachment **41**.

(85) 7. Lock Assembly for Operable Portion **20**

(86) FIG. **15** includes views each showing a lock assembly for the operable portion **20** shown in FIG. **7**. As shown in FIGS. **12** and **15**, the first attachment **41** has the loose-fitting hole **41h** for loosely receiving the locking magnet **27** with the operable portion **20** being engaged with the first attachment **41**.

(87) FIG. **15** includes a view **1201** showing the operable portion **20** not at the position for engagement with the first attachment **41**, or more specifically showing the engagement portion **20a** of the base **23** out of engagement with the tab **41a** on the first attachment **41**. In this case, the locking magnet **27** is in contact with the upper surface of the first attachment **41**, and the operable portion **20** is rotatable about an axis aligned with the pressing direction.

(88) FIG. **15** includes a view **1202** showing the operable portion **20** rotated from the position shown in the view **1201** and engaged with the first attachment **41**. More specifically, the view **1202** shows the engagement portion **20a** of the base **23** at the position for engagement with the tab **41a** on the first attachment **41**.

(89) In the state shown in the view **1202**, the locking magnet **27** in the operable portion **20** is aligned with the loose-fitting hole **41h** in the first attachment **41** and falls into the loose-fitting hole **41h** under its weight. This locks the operable portion **20** in the engaged state and is unrotatable about an axis aligned with the pressing direction. The operable portion **20** is thus unremovable from the first attachment **41** fixed to the transparent support plate **13**.

(90) Unlocking the operable portion **20** in the engaged state will now be described. With the operable portion **20** being engaged with the first attachment **41**, the locking magnet **27** is movable in the pressing direction and in the direction opposite to the pressing direction.

(91) FIG. **16** is a view of the button deck **10** (the transparent support plate **13** and the operable portion **20**) shown in FIG. **2** and a fixture **50** for unlocking the operable portion **20** in the engaged state. FIG. **17** is a view of the fixture **50** attached to the operable portion **20**. FIG. **18** is a cross-sectional view of the fixture **50** attached to the operable portion **20**.

(92) As shown in FIGS. **16** and **17**, the fixture **50** is annular and can cover the component compartment **23a** in the base **23** in the operable portion **20** from above. The fixture **50** is not limited to being annular but may be, for example, circular. The fixture **50** includes an unlocking magnet **51** (a removal magnet) at a position corresponding to a position above the locking magnet **27** when the fixture **50** is attached to the operable portion **20**.

(93) In this state, the unlocking magnet **51** has the direction of magnetization parallel to the pressing direction of the button **21** and has a polarity to attract the locking magnet **27**. When the fixture **50** is attached to the operable portion **20** engaged with the first attachment **41** as shown in FIG. **18**, the locking magnet **27** is attracted by the unlocking magnet **51** and removed from the loose-fitting hole **41h**. This unlocks the operable portion **20** and allows rotation of the operable portion **20** for removal from the first attachment **41** fixed to the transparent support plate **13**.

(94) 8. Effects

(95) As described above, the operable portion **20** with the above structure is difficult to remove from the transparent support plate **13** without the dedicated fixture **50** for unlocking the operable portion **20** in the engaged state. This prevents the operable portion **20** from being removed from the

pushbutton switch **15** by a person other than the manager of the amusement machine **1**.

(96) The pushbutton switch **15** includes, as fully separate portions, the operable portion **20** and the detector **30** that are located on different surfaces, specifically the front and back surfaces, of the transparent support plate **13**. The transparent support plate **13** thus eliminates an opening for receiving the pushbutton switch and avoids fracture due to a decrease in strength.

(97) The above structure eliminates work for producing an opening in the transparent support plate **13** and facilitates attachment of the pushbutton switch **15**. The above structure also reduces restrictions on the attachment position of the pushbutton switch **15**, increasing flexibility in attachment.

(98) A known pushbutton switch, which includes an operable portion **20** and a detector **30** integral with each other, is to have a different thickness for a transparent support plate **13** with a different thickness (or with a different depth of the attachment opening). In contrast, the pushbutton switch with the above structure is attachable to the transparent support plate **13** with a different thickness that allows the detector **30** to detect the pressed state of the button **21**.

(99) In the above structure, the electronic components for detecting the pressed state of the button **21** can be included without being located in the operable portion **20**. More specifically, the multiple LEDs **38** for lighting the pushbutton switch **15** are located in the detector **30**. The operable portion **20** uses no electricity and eliminates, for example, wiring, thus facilitating maintenance. The transparent support plate **13** also eliminates openings for wiring.

(100) In the above structure, the button **21** includes a transparent portion. The detector **30** has an opening or is formed from a transparent material in its area corresponding to the button **21**. This allows the screen of the display **12** to be viewed through the button **21**. The first attachment **41** and the second attachment **43** facilitate attachment of the operable portion **20** and the detector **30** to the transparent support plate **13**.

Second Embodiment

(101) Another embodiment of the present invention will now be described. For ease of explanation, the components having the same functions as the components described in the above embodiment are given the same reference numerals as those components and will not be described.

(102) In a pushbutton switch **15** according to a second embodiment, an operable portion **20** includes multiple locking magnets **27**, and a first attachment **41** has multiple loose-fitting holes **41h** for the respective locking magnets **27**. The other features in the second embodiment are the same as or similar to those in the first embodiment. In the second embodiment, a fixture **50** also includes multiple unlocking magnets **51** for the respective multiple locking magnets **27**.

(103) FIG. **19** is a view of a button deck **10** (a transparent support plate **13** and the operable portion **20**) in the second embodiment and the fixture **50** for unlocking the operable portion **20** in the engaged state. FIG. **19** shows the fixture **50** attached to the operable portion **20**. As illustrated, the fixture **50** includes, for example, four unlocking magnets **51**. The operable portion **20** also includes four locking magnets **27** corresponding to the four unlocking magnets **51**.

(104) In the second embodiment, when the operable portion **20** is engaged with the first attachment **41**, the locking magnets **27** are fitted in the respective loose-fitting holes **41h** to lock the engaged operable portion **20**. When the fixture **50** is attached to the operable portion **20** engaged with the first attachment **41**, the locking magnets **27** are attracted by the respective unlocking magnets **51** and removed from the loose-fitting holes **41h**.

(105) This unlocks the operable portion **20** and allows rotation of the operable portion **20** for removal from the first attachment **41** fixed to the transparent support plate **13**. An operable portion **20** including a single locking magnet **27** may be unlocked and removed with a common magnet without the fixture. In contrast, the operable portion **20** including the multiple locking magnets **27** at appropriate positions is more difficult to unlock without the dedicated fixture.

(106) FIG. **20** is a schematic diagram of the locking magnets **27** in the operable portion **20** and the unlocking magnets **51** in the fixture **50** operable portion **20**, showing their polarity arrangements.

The numbers in the figure indicate positions in the operable portion **20** and in the fixture **50** operable portion **20**.

(107) In the example of FIG. **20**, in the operable portion **20**, the poles of the locking magnets **27** adjacent to the fixture **50** are the N pole, the N pole, the S pole, and the S pole in numerical order. In the fixture **50**, the poles of the unlocking magnets **51** adjacent to the operable portion **20** are the S pole, the S pole, the N pole, and the N pole in numerical order to correspond to the poles in the operable portion **20**. Thus, the locking magnets **27** in the operable portion **20** are arranged to attract the corresponding unlocking magnets **51** in the fixture **50**.

(108) However, the polarity arrangement of the multiple locking magnets **27** is not limited to the example of FIG. **20**. The polarity arrangement may be any one of the arrangements below, where a first pole is the S pole or the N pole, and a second pole is the pole opposite to the first pole. The arrangement may be, in numerical order, the first pole, the first pole, the first pole, and the first pole; the first pole, the first pole, the first pole, and the second pole; the first pole, the first pole, the second pole, and the second pole; and the first pole, the second pole, the first pole, and the second pole. Similarly to the above example, the unlocking magnets **51** in the fixture **50** are orientated to have polarities to magnetically attract the corresponding locking magnets **27** for the corresponding numbers.

(109) Among the above arrangements, the second, third, or fourth arrangement combines different polarities and may allow more difficult unlocking with a single magnet. In other words, at least one of the multiple movable magnets may be oriented to have a polarity different from the polarity of at least another one of the multiple movable magnets.

(110) Overview

(111) A pushbutton switch according to an aspect of the present invention includes an attachment fixed to one surface of a support being a plate, an operable portion removably engaged with the attachment, and a detector attached to another surface of the support and facing the operable portion. The operable portion includes a press portion and a movable magnet movable in a pressing direction from the operable portion toward the detector and in a direction opposite to the pressing direction. The operable portion is rotatable about an axis aligned with the pressing direction to be engaged with the attachment. The attachment has a loose-fitting hole to loosely receive the movable magnet with the operable portion being engaged with the attachment.

(112) The above structure prevents the operable portion from being removed from the pushbutton switch by a person other than the manager. The operable portion in the engaged state is locked using the movable magnet and can be unlocked by the manager.

(113) In the pushbutton switch according to the above aspect, the attachment may include a tab. The operable portion may include an engagement portion. The operable portion may be engaged with the attachment through engagement of the engagement portion with the tab. The above structure can securely engage the operable portion with the attachment through engagement of the engagement portion with the tab.

(114) In the pushbutton switch according to the above aspect, the operable portion may further include a base supporting the press portion. The base may include the engagement portion. The above structure can tightly fasten the operable portion to the attachment through engagement of the base supporting the press portion with the attachment.

(115) A pushbutton switch unit according to an aspect of the present invention includes the pushbutton switch according to the above aspect, and a fixture detachably attachable to the operable portion. The fixture includes a removal magnet that magnetically removes the movable magnet from the loose-fitting hole when the fixture is attached to the operable portion engaged with the attachment.

(116) The above structure prevents the operable portion from being removed from the pushbutton switch by a person other than the manager. The operable portion can be unlocked by the manager using the fixture.

(117) In the pushbutton switch according to the above aspect, the operable portion may include a plurality of the movable magnets. The attachment may have a plurality of the loose-fitting holes for the respective plurality of movable magnets. Further, at least one of the plurality of movable magnets may be oriented to have a polarity different from a polarity of at least another one of the plurality of movable magnets. The above structure more effectively prevents the operable portion from being removed from the pushbutton switch by a person other than the manager. The operable portion can be unlocked by the manager using the fixture.

(118) A pushbutton switch unit according to an aspect of the present invention includes the pushbutton switch according to the above aspect, and a fixture detachably attachable to the operable portion. The fixture includes a plurality of removal magnets for the respective plurality of movable magnets. The plurality of removal magnets magnetically remove the respective plurality of movable magnets from the respective plurality of loose-fitting holes when the fixture is attached to the operable portion engaged with the attachment. The above structure more effectively prevents the operable portion from being removed from the pushbutton switch by a person other than the manager.

(119) An operating unit according to an aspect of the present invention includes a transparent support and a display together serving as a touchscreen, and a pushbutton switch. The pushbutton switch includes an attachment fixed to a surface of the transparent support opposite to the display, an operable portion removably engaged with the attachment, and a detector attached to a surface of the transparent support adjacent to the display. The detector faces the operable portion. The operable portion includes a press portion and a movable magnet movable in a pressing direction from the operable portion toward the detector and in a direction opposite to the pressing direction. The operable portion is rotatable about an axis aligned with the pressing direction to be engaged with the attachment. The attachment has a loose-fitting hole to loosely receive the movable magnet with the operable portion being engaged with the attachment.

(120) The above structure prevents the operable portion from being removed from the pushbutton switch by a person other than the manager. The operable portion in the engaged state is locked using the movable magnet and can be unlocked by the manager.

(121) An amusement machine according to an aspect of the present invention includes a first display that displays an image for amusement, a transparent support and a second display together serving as a touchscreen, and a pushbutton switch. The pushbutton switch includes an attachment fixed to a surface of the transparent support opposite to the second display, an operable portion removably engaged with the attachment, and a detector attached to a surface of the transparent support adjacent to the second display. The detector faces the operable portion. The operable portion includes a press portion and a movable magnet movable in a pressing direction from the operable portion toward the detector and in a direction opposite to the pressing direction. The operable portion is rotatable about an axis aligned with the pressing direction to be engaged with the attachment. The attachment has a loose-fitting hole to loosely receive the movable magnet with the operable portion being engaged with the attachment.

(122) The above structure prevents the operable portion from being removed from the pushbutton switch by a person other than the manager. The operable portion in the engaged state is locked using the movable magnet and can be unlocked by the manager.

(123) The present invention is not limited to the above embodiments, but may be modified variously within the spirit and scope of the claimed invention. The technical means described in different embodiments may be combined as appropriate in other embodiments within the technical scope of the invention. The technical means described in different embodiments may be combined to produce a new technical feature.

Claims

1. A pushbutton switch, comprising: an attachment fixed to a first surface of a support comprising a plate; an operable portion removably engaged with the attachment; and a detector attached to a second surface of the support and facing the operable portion, wherein the operable portion comprises a press portion and a movable magnet movable in a pressing direction from the operable portion toward the detector and in a direction opposite to the pressing direction, and the operable portion is rotatable about an axis aligned with the pressing direction to be engaged with the attachment, the attachment comprises a loose-fitting hole to loosely receive the movable magnet with the operable portion being engaged with the attachment, the movable magnet of the operable portion comprises a plurality of movable magnets, and the loose-fitting hole of the attachment comprises a plurality of loose-fitting holes for the respective plurality of movable magnets.
2. The pushbutton switch according to claim 1, wherein the attachment comprises a tab, the operable portion comprises an engagement portion, and the operable portion is engaged with the attachment through engagement of the engagement portion with the tab.
3. A pushbutton switch unit, comprising: the pushbutton switch according to claim 2; and a fixture detachably attachable to the operable portion, wherein the fixture comprises a plurality of removal magnets for the respective plurality of movable magnets, and the plurality of removal magnets magnetically remove the respective plurality of movable magnets from the respective plurality of loose-fitting holes when the fixture is attached to the operable portion engaged with the attachment.
4. The pushbutton switch according to claim 2, wherein the operable portion further comprises a base supporting the press portion, and the base comprises the engagement portion.
5. A pushbutton switch unit, comprising: the pushbutton switch according to claim 4; and a fixture detachably attachable to the operable portion, wherein the fixture comprises a plurality of removal magnets for the respective plurality of movable magnets, and the plurality of removal magnets magnetically remove the respective plurality of movable magnets from the respective plurality of loose-fitting holes when the fixture is attached to the operable portion engaged with the attachment.
6. The pushbutton switch according to claim 1, wherein at least one of the plurality of movable magnets is oriented to have a polarity different from a polarity of at least another one of the plurality of movable magnets.
7. A pushbutton switch unit, comprising: the pushbutton switch according to claim 6; and a fixture detachably attachable to the operable portion, wherein the fixture comprises a plurality of removal magnets for the respective plurality of movable magnets, and the plurality of removal magnets magnetically remove the respective plurality of movable magnets from the respective plurality of loose-fitting holes when the fixture is attached to the operable portion engaged with the attachment.
8. A pushbutton switch unit, comprising: the pushbutton switch according to claim 1; and a fixture detachably attachable to the operable portion, wherein the fixture comprises a plurality of removal magnets for the respective plurality of movable magnets, and the plurality of removal magnets magnetically remove the respective plurality of movable magnets from the respective plurality of loose-fitting holes when the fixture is attached to the operable portion engaged with the attachment.
9. A pushbutton switch unit, comprising: a pushbutton switch; and a fixture, wherein the pushbutton switch comprises: an attachment fixed to a first surface of a support comprising a plate; an operable portion removably engaged with the attachment; and a detector attached to a second surface of the support and facing the operable portion, the operable portion comprises a press portion and a movable magnet movable in a pressing direction from the operable portion toward the detector and in a direction opposite to the pressing direction, and the operable portion is rotatable about an axis aligned with the pressing direction to be engaged with the attachment, the attachment comprises a loose-fitting hole to loosely receive the movable magnet with the operable portion being engaged with the attachment, the fixture is detachably attachable to the operable portion, and the fixture comprises a removal magnet configured to magnetically remove the movable magnet from the loose-fitting hole when the fixture is attached to the operable portion

engaged with the attachment.

10. An operating unit, comprising: a transparent support and a display together serving as a touchscreen; and a pushbutton switch, wherein the pushbutton switch comprises an attachment fixed to a surface of the transparent support opposite to the display, an operable portion removably engaged with the attachment, and a detector attached to a surface of the transparent support adjacent to the display, the detector facing the operable portion, the operable portion comprises a press portion and a movable magnet movable in a pressing direction from the operable portion toward the detector and in a direction opposite to the pressing direction, and the operable portion is rotatable about an axis aligned with the pressing direction to be engaged with the attachment, the attachment has a loose-fitting hole to loosely receive the movable magnet with the operable portion being engaged with the attachment, the movable magnet of the operable portion comprises a plurality of movable magnets, and the loose-fitting hole of the attachment comprises a plurality of loose-fitting holes for the respective plurality of movable magnets.

11. An amusement machine, comprising: a first display configured to display an image for amusement; a transparent support and a second display together serving as a touchscreen; and a pushbutton switch, wherein the pushbutton switch comprises an attachment fixed to a surface of the transparent support opposite to the second display, an operable portion removably engaged with the attachment, and a detector attached to a surface of the transparent support adjacent to the second display, the detector facing the operable portion, the operable portion comprises a press portion and a movable magnet movable in a pressing direction from the operable portion toward the detector and in a direction opposite to the pressing direction, and the operable portion is rotatable about an axis aligned with the pressing direction to be engaged with the attachment, the attachment has a loose-fitting hole to loosely receive the movable magnet with the operable portion being engaged with the attachment, the movable magnet of the operable portion comprises a plurality of movable magnets, and the loose-fitting hole of the attachment comprises a plurality of loose-fitting holes for the respective plurality of movable magnets.

12. A pushbutton switch unit, comprising: a pushbutton switch; and a fixture, wherein the pushbutton switch comprises: an attachment fixed to a first surface of a support comprising a plate; an operable portion removably engaged with the attachment; and a detector attached to a second surface of the support and facing the operable portion, wherein the operable portion comprises a press portion and a movable magnet movable in a pressing direction from the operable portion toward the detector and in a direction opposite to the pressing direction, and the operable portion is rotatable about an axis aligned with the pressing direction to be engaged with the attachment, the attachment comprises a loose-fitting hole to loosely receive the movable magnet with the operable portion being engaged with the attachment, the attachment comprises a tab, the operable portion comprises an engagement portion, the operable portion is engaged with the attachment through engagement of the engagement portion with the tab, the fixture is detachably attachable to the operable portion, and the fixture comprises a removal magnet configured to magnetically remove the movable magnet from the loose-fitting hole when the fixture is attached to the operable portion engaged with the attachment.

13. A pushbutton switch unit, comprising: a pushbutton switch; and a fixture, wherein the pushbutton switch comprises: an attachment fixed to a first surface of a support comprising a plate; an operable portion removably engaged with the attachment; and a detector attached to a second surface of the support and facing the operable portion, wherein the operable portion comprises a press portion and a movable magnet movable in a pressing direction from the operable portion toward the detector and in a direction opposite to the pressing direction, and the operable portion is rotatable about an axis aligned with the pressing direction to be engaged with the attachment, the attachment comprises a loose-fitting hole to loosely receive the movable magnet with the operable portion being engaged with the attachment, the attachment comprises a tab, the operable portion comprises an engagement portion, the operable portion is engaged with the attachment through

engagement of the engagement portion with the tab, the operable portion further comprises a base supporting the press portion, and the base comprises the engagement portion, the fixture is detachably attachable to the operable portion, and the fixture comprises a removal magnet configured to magnetically remove the movable magnet from the loose-fitting hole when the fixture is attached to the operable portion engaged with the attachment.
